# Seihwan Jung

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## Education

**Seoul National University (SNU)** Mar. 2024 – Feb. 2028 (expected)
*B.S. in Mechanical Engineering* *Seoul, South Korea*

- GPA: 3.55/4.3

**Technische Universität Berlin (TU Berlin)** Oct. 2026 – Mar. 2027
*Exchange Student* *Berlin, Germany*

- Winter Semester 2026/27; continuing coursework and research in robotic manipulation.

## Research Interests

- **Grasping and Manipulation:** Passive compliance and adaptive grasp stability – letting a mechanism conform to an object instead of controlling every joint.
- **Robotic Hardware Design:** Underactuated, tendon-driven mechanisms, from CAD through fabrication and assembly to tested hardware.
- **Simulation-to-Real:** Using physics simulation sweeps to choose design parameters, then verifying against measurements on the physical build.

## Experience

**SNU BioRobotics Laboratory (BRL)** Jan. 2026 – Aug. 2026
*Undergraduate Researcher* *Seoul, South Korea*

- Designed, fabricated, and validated a tendon-driven prosthetic hand with a fully passive abduction/adduction (ADD/ABD) degree of freedom.
- Full pipeline ownership: mechanical design, CAD (Fusion 360), MuJoCo simulation sweeps, 3D-printed fabrication, and physical testing.

## Selected Projects (see more at seihwan-jung.github.io)

**Passive-ABD Tendon-Driven Prosthetic Hand** More Info.

- Fully passive ADD/ABD joint at the MCP using two opposing torsion springs and dual-tendon routing – no extra motors or added weight.
- Selected spring stiffness via a MuJoCo simulation sweep across object families, then implemented physical control with Dynamixel actuators.

## Honors / Awards

**2nd Place – Creative Engineering Design Robocon** *Nov. 2025*
Department of Mechanical Engineering 33rd Anniversary, Seoul National University

## Skills

- **Programming:** Python, C++, MATLAB
- **Hardware:** CAD (Fusion 360), 3D Printing, Motor Control (Dynamixel), Microcontrollers (Arduino, OpenRB, ESP-32), Workshop Fabrication
- **Simulation:** MuJoCo
- **Languages:** Korean (Native), English (Working Proficiency)